Introduction to Artificial Intelligence
Assignment 2: Creative Image Interpreter

Deadline

Week 15 (01.12.25 23:59)

Submission

1. Only one source code file has to be uploaded to Moodle.
2. Report has to be uploaded to Moodle with the file name like *NameSurname.pdf,* e.g. *IvanIvanov.pdf*. No other symbols allowed. A PDF-report of 2-3 pages in IEEE format which gives a description of your method and talks about the design process and testing used in order to make your player agent. You can find the template files at: (https://www.ieee.org/conferences/publishing/templates.html).
3. For each test image 3 best generated output JPG-files are expected, where each of them should be received from different run. File names should be like *NameSurnameOutputX_Y_Z.jpg,* e.g. *IvanIvanovOutput1_1_1.jpg* (more in Outputs section)

Programming Languages

Java (JDK 8 or 21) or C++ (17 or 20 or 23) or C# (8 or 10 or 13) or Python 3

Requests

- The program must work, the code should be readable, well-structured and should contain English comments in a language-specified format
- It has to be only one code file
- It has to be only one report **.pdf* file
- It is NOT allowed to use any online APIs, LLMs and libraries with built-in EA algorithms
- NO extension of a deadline. Works sent after the deadline will NOT be evaluated
- Excused students can have extension of deadline by number of excused days intersecting with assignment days based on DoE's information. Still, the deadline can't be pushed later than 04/12/25 12 AM (noon).
- Assignment is individual
- We will be using MOSS (Measure of Software Similarity) as a test for plagiarism. Be reminded that a score of 0 will be assigned to any submissions suspected of plagiarism pending a full investigation as per IU policies

Grading Criteria

- 15% result distinguishability (if not similar to original result, then result creativity is 0)
- 20% result creativity
- 20% for live grading (if results are significantly worse, then the result distinguishability, result creativity and ranking are 0)
- 10% for code readability and comments quality
- 5% for ranking in the art contest among teachers
- 30% for the well-structured and informative report

Task

This assignment has a project nature and requires you to express your creativity. You will be given several input images of size 512x512 pixels. For each of them via any evolutionary algorithm you will need to produce 3 similar 512x512 images which will be presented to your classmates from all groups. The algorithm should be able to finish work at most in 10 min on your own computer for any image, after which the 3 best results have to be generated.

There is no single solution, results may significantly vary. Generated result should not be equivalent to the source image, but has to be somehow similar to the original. Impossibility to detect the source from the obtained result indicates invalid solution. You are asked to make something similar but with your creative touch. Examples may include, but not limited to evolving colored pixels, geometric shapes, combined smaller images, mosaics, etc. You are obliged to use both crossover and mutation for evolving your images.

Report

Consider several sections to make your report mature and structured. Your report should deeply describe chosen EA algorithm, its flow, chromosome representation, fitness function, crossover, mutation, population size, stopping criteria and other parameters in plain English. The explanation of your methodology has to be clear to make it possible to reproduce your idea by readers.

Provide statistics supported by tables about held experiments. For this you are expected to run the same code many times for each test image and find the average fitness values for the last generations, average time used by algorithm, the average number of generations, etc. Ideally, statistics should demonstrate that your algorithm is more or less stable and finds solutions in most of the runs. If your algorithm finds solutions rarely, then it's not reliable and requires updates, use it as a feedback. You are free to use your own extra tests for statistics, assume that your algorithms will be tested on other images after submission. Statistics have to be thoroughly analyzed and discussed.

For each tested image the 3 plots demonstrating the average population fitness and maximum fitness values among generations have to be presented (X-axis is for fitness value, Y-axis for the number of generations). Each plot has to represent one separate code run for the specific image. Use different colors for curves. In total, it is expected to have 3 plots for each tested image. Analysis and comparison of plots also has to be presented in the report.

References have to be provided, if any were used. Also, if any extra libraries, dependencies were used, it has to be explicitly mentioned, what was used and for which purpose. In case of not using any extra tools, this also has to be stated.

Evaluation

Convergence of EAs for the same task may be varying in terms of time and generations. It depends on implementation, hardware and software, including programming language. However, we assume that 10 min should be enough for any student. On 4th of December, we plan to provide live grading with hidden tests. For this you will need to bring your laptops. This procedure is required to verify that your code is working since running your codes on our computers may easily lead to errors, especially for machine-dependent languages. Consider that low results of live grading may negatively affect other parts of the grade.

Inputs

The inputs are represented by *inputX.jpg* file, where *X* is the index of the image. Example of the input is provided on Figure 1. You are guaranteed to have image indices starting from 1 and incrementing by 1.

Figure 1. Input1 example

Outputs

The output for *inputX.jpg* is represented by 5 files called *NameSurnameOutputX_Y_Z.jpg*, where *X* is the index of the image, *Y* is the number of run for image from 1 to 3, *Z* is the value from 1 to 5, indicating the index among the best results per run. Do not get confused 5 output files per run per image with submission of 3 images per test image to Moodle. The idea is that from each run you are generating 5 best outcomes, but only 1 of them of your choice can be submitted. The rest 4 is for live grading to be safe and have some variety. Assume that best results are not obliged to be from the last generation only. Consider possible outcomes taken from Internet on Figure 2.

You are not supposed to do different versions of results, one is more than enough. The Figure 2 demonstrates that the achieved result may be creative and still resemble the original image. Keep in mind that for the submission you are supposed to take only one best image per run. In total at least 3 runs are expected to get 3 images for submission.

Figure 2. Possible output examples